



Triductor TR8253/TR8253L EtherCAT Slave Chipset Datasheet

version 2.00



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Revision History

Version	Date (mm-dd-yy)	Author	Reason
1.01	12-12-23	Kai	preliminary version creation
1.02	01-30-24	Kai	Fix some Pin number type errors in Table12,Table13
1.03	10-08-24	Szhou	Fix Pin64 of TR8253 to NC
2.00	12-08-24	Szhou	Add PDI QSPI mode and HBI Index mode

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1. Overview

Triductor's TR8253/TR8253L is a series of EtherCAT Slave Controller chips, which integrate ESC core module licensed from Beckhoff Automation.

It supports 2/3 ports of the EtherCAT communication for the EtherCAT slave fieldbus application. It is suitable for industrial automation, motor control, motion control, robotics, digital signal I/O control, sensor data acquisition, and other realtime industrial control product applications.

TR8253/TR8253L provides two process data interfaces, i.e. μ Controller Local Bus interface and SPI Slave interface, which can be used to connect the TR8253/TR8253L to external MCU/DSP to support EtherCAT functionality.

TR8253/TR8253L supports a wide range of applications. For example, it may be used as a 16 bit Digital I/O node without external logic using Distributed clocks, or as a part of a complex μ Controller design with up to three EtherCAT communication ports.

TR8253/TR8253L also integrates two fast Ethernet PHYs that support 100Mbps full duplex operation and HP Auto - MDIX functionality.

1.1. Block Diagram

The general functionality of the EtherCAT Slave Controller (ESC) is shown in Figure 1:

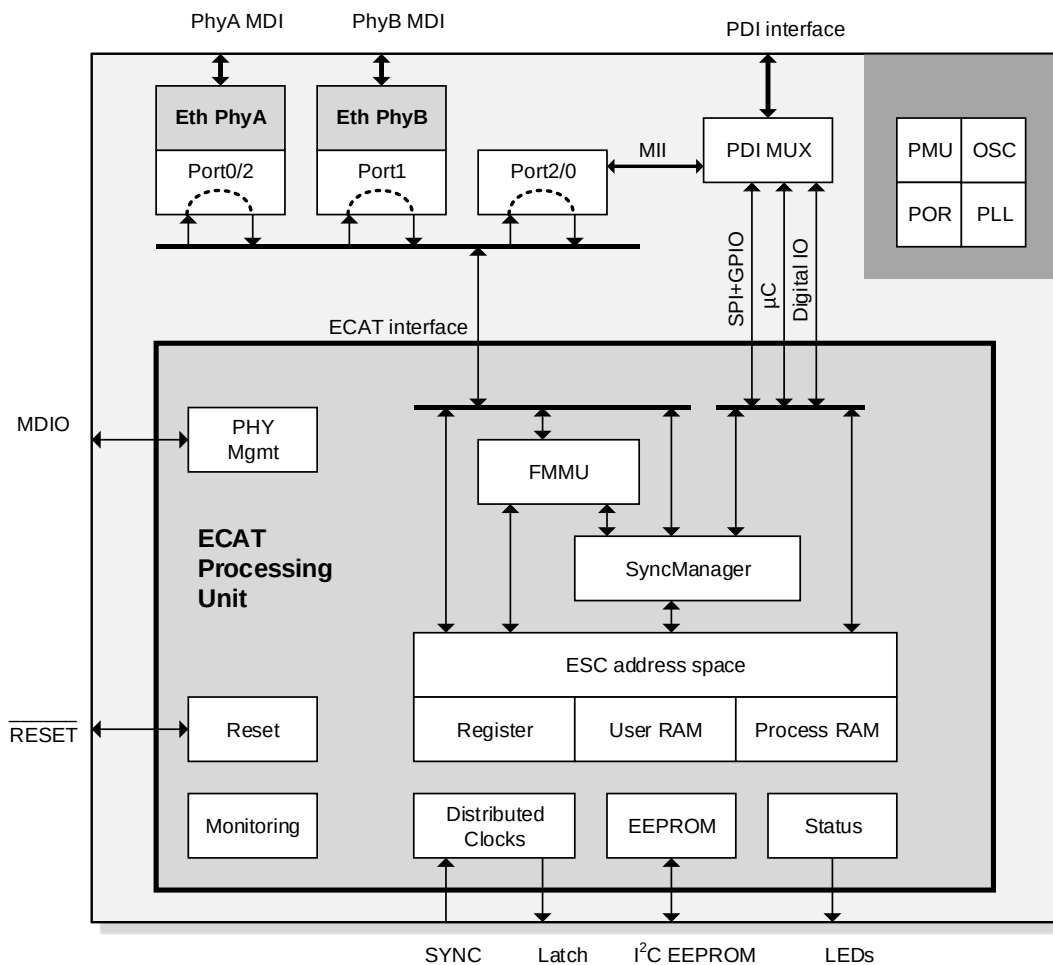


Figure 1 System Block Diagram

1.2. Features

EtherCAT Slave Controller functionality

2-3 EtherCAT communication Ports
 8x Fieldbus Memory Management Units (FMMUs)
 8x Sync Managers
 64 - bit distributed clock
 8K Bytes Process RAM
 128 Bytes User RAM

Process Data Interfaces

16/32 Bit Digital I/O (unidirectional/bidirectional)
 SPI Slave in direct addressing mode
 8/16bit multiplexed μ Controller interface

Fast Ethernet PHY

Compatible with IEEE802.3/802.3u 100BASE - TX standards
 Support for automatic detection and switching of cross lines (HP Auto MDIX)
 Supports automatic polarity detection and correction

Power supply

Supports single 3.3V Power supply
 Integrated voltage regulator for logic core/PLL/PHY

Operating Temperature Range

- 40 to +85°C

Table 1 TR8253/ TR8253L Features

Symbol	TR8253	TR8253L
Ports	2-3 ports MII	2-3 ports MII
FMMUs	8	8
Sync Managers	8	8
RAM	8 Kbyte	8 Kbyte
Distributed Clocks	64 bit	64 bit
	16 Bit Digital I/O	16 Bit Digital I/O
Process Data Interfaces	SPI/SQI Slave (direct addressing)	SPI/SQI Slave (direct addressing)
	8/16 Bit Host bus interface	8/16 Bit Host bus interface
Power supply	Integrated voltage regulator for logic core/PLL	Integrated voltage regulator for logic core/PLL
I/O	3.3V compatible I/O	3.3V compatible I/O
Package	QFN64,9x9mm ² ,0.5mm pitch	LGA64,9x9mm ² ,0.5mm pitch

1.3. Applications

- Motor Motion Control
- Digital signal I/O control
- Process/Factory Automation
- Communication Modules, Interface Cards
- Sensors Data Acquisition
- Operator HMI Interfaces

2. Pin Assignments

2.1. Overview

These conventions are used for the control/status signal name:

SIGNAL active HIGH

SIGNAL# active LOW

These conventions are used for the signal type:

I	Input
I _{PU}	Input with internal pull-up
I _{PD}	Input with internal pull-down
I _{CLK}	Input clock
O	Output
O _{3S}	state output
O _{OD}	Open-drain output
O _{CLK}	Output clock
IO	Bidirectional
IO _{3S}	state IO
I _A	Analog Input
O _A	Analog Output
IO _A	Analog IO
PWR	Power supply
PWR _A	Analog power supply
GND	Ground
GND _A	Analog ground
NC	Not connect, leave OPEN

2.2. TR8253 Pin Assignment

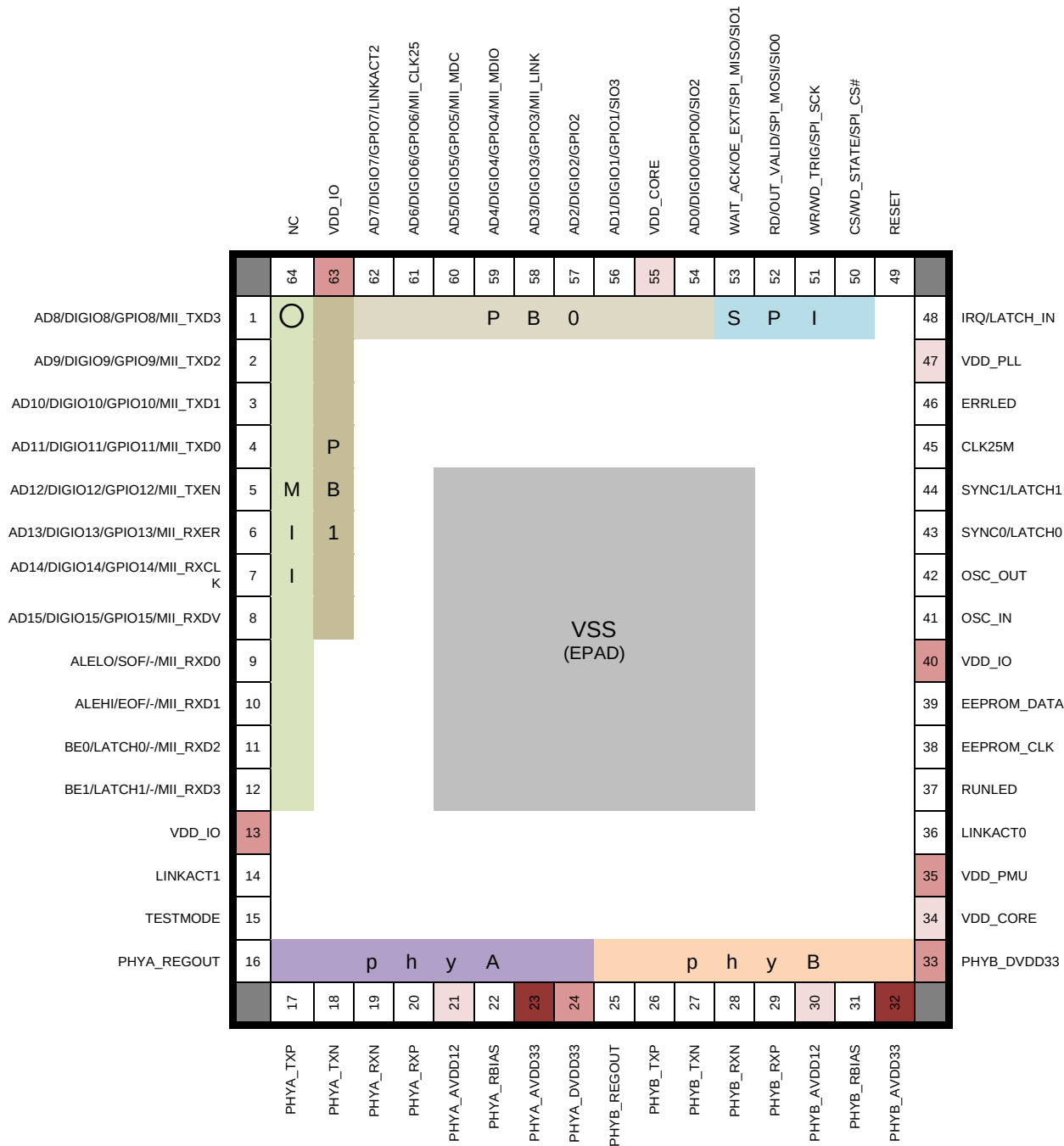


Figure 2 TR8253 Pin Assignment – Top View

Table 2 TR8253 Pin Assignment

Pin	PDI MODE					Config	Type
	HBI Index	HBI Multiplex	SPI/SQI+GPIO	SPI/SQI+MII	Digital IO		
1	D8	AD8	GPIO8	MII_TXD3	DIGIO8	TX_SHI[1]	IO
2	D9	AD9	GPIO9	MII_TXD2	DIGIO9	TX_SHI[0]	IO
3	D10	AD10	GPIO10	MII_TXD1	DIGIO10		IO
4	D11	AD11	GPIO11	MII_TXD0	DIGIO11		IO
5	D12	AD12	GPIO12	MII_TXEN	DIGIO12		IO
6	D13	AD13	GPIO13	MII_RXER	DIGIO13		IO
7	D14	AD14	GPIO14	MII_RXCLK	DIGIO14		IO
8	D15/A0	AD15	GPIO15	MII_RXDV	DIGIO15		IO
9	A1	ALELO	SIO2	MII_RXD0	SOF		IO
10	A2	ALEHI	SIO3	MII_RXD1	EOF		IO
11	A3	BE0	nc ⁽¹⁾	MII_RXD2	LATCH0		IO
12	A4	BE1	nc ⁽¹⁾	MII_RXD3	LATCH1		IO
13	VDD_IO						PWR _A
14	LINKACT1					MODE[1] LEDPOL1	IO _{PD}
15	TESTMODE						I _{PD}
16	PHYA_REGOUT						PWR _A
17	PHYA_TXP						IO _A
18	PHYA_TXN						IO _A
19	PHYA_RXN						IO _A
20	PHYA_RXP						IO _A
21	PHYA_AVDD12						PWR _A
22	PHYA_RBIAS						IO _A
23	PHYA_AVDD33						PWR _A
24	PHYA_DVDD33						PWR _A
25	PHYB_REGOUT						PWR _A
26	PHYB_TXP						IO _A
27	PHYB_TXN						IO _A
28	PHYB_RXN						IO _A
29	PHYB_RXP						IO _A
30	PHYB_AVDD12						PWR _A
31	PHYB_RBIAS						IO _A
32	PHYB_AVDD33						PWR _A
33	PHYB_DVDD33						PWR _A
34	VDD_CORE						PWR _A
35	VDD_PMU						PWR _A
36	LINKACT0					MODE[0] LEDPOL0	IO _{PD}
37	RUNLED/STATE_RUNLED					EEPROM_SIZE LEDPOL3	IO _{PD}
38	EEPROM_CLK						IO _{PU}
39	EEPROM_DATA						IO _{PU}

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40	VDD_IO						PWR _A
41	OSC_IN						O _A
42	OSC_OUT						IO
43	SYNC0/LATCH0						IO
44	SYNC1/LATCH1						IO
45	CLK25M					CLK25_EN	IO _{PU}
46	ERRLED					LEDPOL4	IO
47	VDD_PLL						PWR _A
48	IRQ			LATCH_IN			IO _{PD}
49	RESET#						I _{PU}
50	CS		SPI_CS#		WD_STATE		IO
51	WR/ENB		SPI_SCK		WD_TRIG		IO
52	RD/WR_RD		SPI_MOSI/SIO0		OUT_VALID		IO
53	WAIT_ACK		SPI_MISO/SIO1		OE_EXT		IO
54	D0	AD0	GPIO0	GPIO0/SIO2	DIGIO0		IO
55	VDD_CORE						PWR _A
56	D1	AD1	GPIO1	GPIO1/SIO3	DIGIO1		IO
57	D2	AD2	GPIO2	nc ⁽¹⁾	DIGIO2		IO
58	D3	AD3	GPIO3	MII_LINK	DIGIO3		IO
59	D4	AD4	GPIO4	MII_MDIO	DIGIO4		IO
60	D5	AD5	GPIO5	MII_MDC	DIGIO5		IO
61	D6	AD6	GPIO6	MII_CLK25	DIGIO6		IO
62	D7	AD7	GPIO7	LINKACT2	DIGIO7	LINKPOL LEDPOL2	IO
63	VDD_IO						PWR _A
64	nc ⁽¹⁾						IO
EPAD	VSS						GND

Note1: PDI pins in 'NC' state should be left open on PCB board.

2.3. TR8253L Pin Assignment

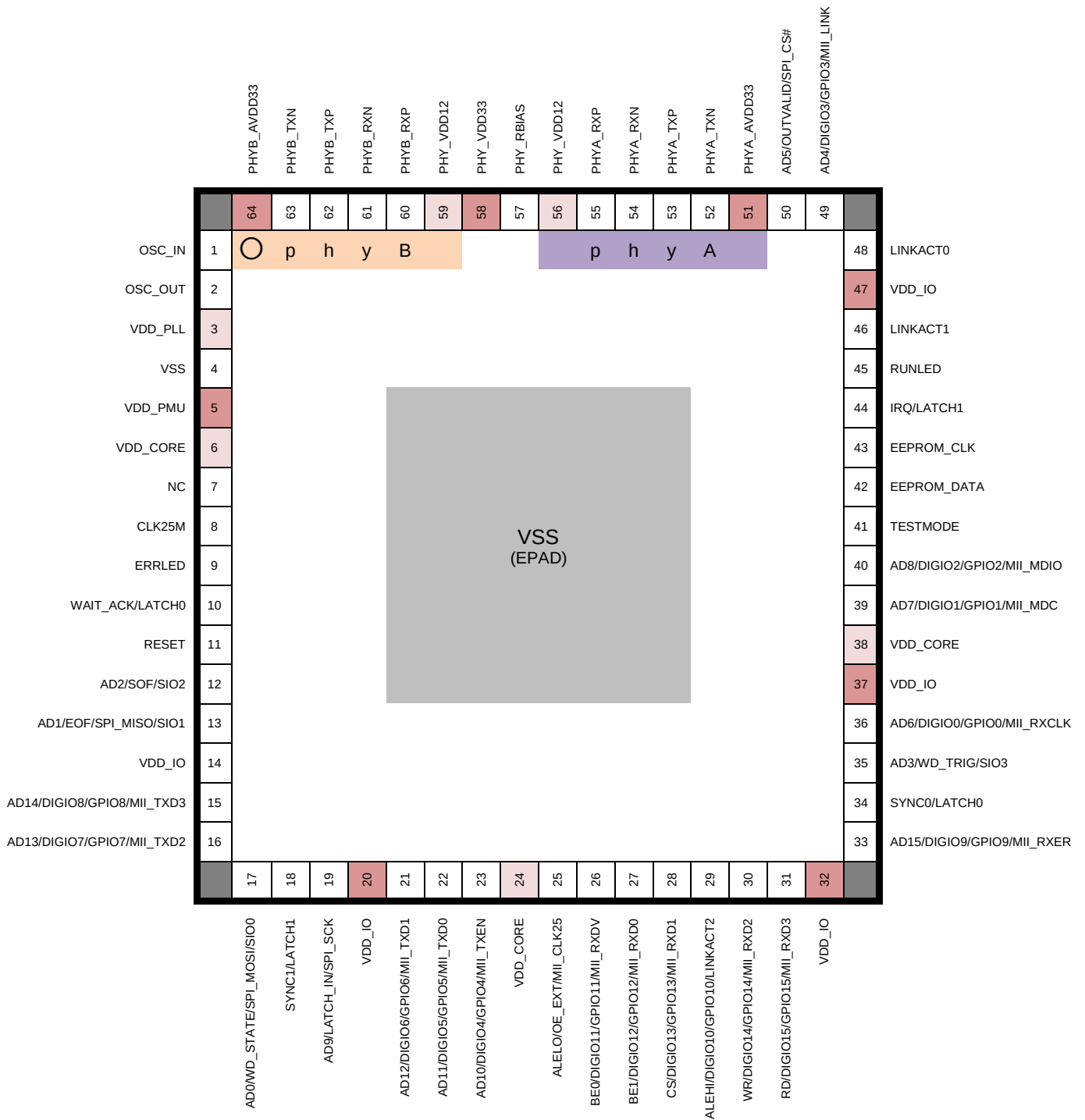


Figure 3 TR8253L Pin Assignment Diagram – Top View

Table 3 TR8253L Pin Assignment

Pin	PDI MODE					Config	Type
	HBI Index	HBI Multiplex	SPI/SQI+GPIO	SPI/SQI+MII	Digital IO		
1			OSC_IN				I _A
2			OSC_OUT				O _A
3			VDD_PLL				PWR _A
4			VSS				GND
5			VDD_PMU				PWR _A
6			VDD_CORE				PWR
7			nc ⁽¹⁾				IO
8			CLK25M			CLK25_EN	IO _{PU}
9			ERRLED			LEDPOL4	IO
10			WAIT_ACK		LATCH0		IO
11			RESET#				I _{PU}
12	D2	AD2	SIO2		SOF		IO
13	D1	AD1	SPI_MISO/SIO1		EOF		IO
14			VDD_IO				PWR _A
15	D14	AD14	GPIO8	MII_TXD3	DIGIO8	TX_SHI[1]	IO
16	D13	AD13	GPIO7	MII_TXD2	DIGIO7	TX_SHI[0]	IO
17	D0	AD0	SPI_MOSI/SIO0		WD_STATE		IO
18			SYNC1/LATCH1				IO
19	D9	AD9	SPI_SCK		LATCH_IN		IO
20			VDD_IO				PWR _A
21	D12	AD12	GPIO6	MII_TXD1	DIGIO6		IO
22	D11	AD11	GPIO5	MII_TXD0	DIGIO5		IO
23	D10	AD10	GPIO4	MII_TXEN	DIGIO4		IO
24			VDD_CORE				PWR
25	A1	ALELO	-	MII_CLK25	OE_EXT		IO
26	A3	BE0	GPIO11	MII_RXDV	DIGIO11		IO
27	A4	BE1	GPIO12	MII_RXD0	DIGIO12		IO
28		CS	GPIO13	MII_RXD1	DIGIO13		IO
29	A2	ALEHI	GPIO10	LINKACT2	DIGIO10	LINKPOL LEDPOL2	IO
30		WR/ENB	GPIO14	MII_RXD2	DIGIO14		IO
31		RD/RD_WR	GPIO15	MII_RXD3	DIGIO15		IO
32			VDD_IO				PWR _A
33	A0/D15	AD15	GPIO9	MII_RXER	DIGIO9		IO
34			SYNC0/LATCH0				IO
35	D3	AD3	SIO3		WD_TRIG		IO
36	D6	AD6	GPIO0	MII_RXCLK	DIGIO0		IO
37			VDD_IO				PWR _A
38			VDD_CORE				PWR
39	D7	AD7	GPIO1	MII_MDC	DIGIO1		IO
40	D8	AD8	GPIO2	MII_MDIO	DIGIO2		IO

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41	TESTMODE						I _{PD}
42	EEPROM_DATA						IO _{PU}
43	EEPROM_CLK						IO _{PU}
44	IRQ			LATCH1			IO
45	RUNLED/STATE_RUNLED					EEPROM_SIZE LEDPOL3	IO _{PD}
46	LINKACT1					MODE[1] LEDPOL1	IO _{PD}
47	VDD_IO						PWR _A
48	LINKACT0					MODE[0] LEDPOL0	IO _{PD}
49	D4	AD4	GPIO3	MII_LINK	DIGIO3		IO
50	D5	AD5	SPI_CS#	SPI_CS#	OUTVALID		IO
51	PHYA_AVDD33						PWR _A
52	PHYA_TXN						IO _A
53	PHYA_TXP						IO _A
54	PHYA_RXN						IO _A
55	PHYA_RXP						IO _A
56	PHY_VDD12						PWR _A
57	PHY_RBIA5						IO _A
58	PHY_VDD33						PWR _A
59	PHY_VDD12						PWR _A
60	PHYB_RXP						IO _A
61	PHYB_RXN						IO _A
62	PHYB_TXP						IO _A
63	PHYB_TXN						IO _A
64	PHYB_AVDD33						PWR _A
EPA D	VSS						GND

Note1: PIN 7 in 'nc' state should be left open or connected to VDD_IO on PCB board.

2.4. Configuration Pins

The configuration pins are used to configure the TR8253/TR8253L at power-on with pull-up or pull-down resistors. At power-on the TR8253/TR8253L uses these pins as inputs to latch the configuration. After power on, the pins have their operation functionality which has been assigned to them, and therefore pin direction changes if necessary. The power-on phase finishes before the RESET# pin is released. In subsequent reset phases without power-on condition, the configuration pins still have their operation functionality, i.e., the TR8253/TR8253L configuration is not latched again and output drivers remain active. The configuration value 0 is realized by a pull-down resistor, a pull-up resistor is used for a 1. Since some configuration pins are also used as LED outputs, the polarity of the LED output depends on the configuration value.

2.4.1. Chip Mode and Port Configuration

Port MODE[1:0] configures the number of physical ports and the mode of PDI pins. It is shown in Table below

Table 4 Chip Mode and Port Configuration

TR8253	TR8253L	Config signal	Register	Description
36	48	MODE[0]	0x0E02[0]	Port Mode 0x = 2 ports mode 10 = 3 ports downstream mode 11 = 3 ports upstream mode
14	46	MODE[1]	0x0E02[1]	

For configurations with 2 ports, logical ports 0 and 1 are used. The port signals are available at physical ports 0 and 1, depending on the port configuration. MODE[1] has to be set to 0.

For configurations with 3 ports, MODE[1] has to be set to 1. MODE[0] determines downstream or upstream application.

For downstream application, phyA is mapped to logical port0 and MII is mapped to logical port2. For upstream application, phyA is mapped to logical port2 and MII is mapped to logical port0. phyB is always mapped to logical port1 for both applications.

Table 5 Chip Mode and Port Mapping

MODE[1:0]	Physical Port		
	MII	PhyB	PhyA
0x	PDI	Logical port 1	Logical port 0
10	Logical port 2	Logical port 1	Logical port 0
11	Logical port 0	Logical port 1	Logical port 2

Once the mode of the chip is selected, the Process Data Interface (PDI) in use is selected by the Process Data Interface (PDI_SEL) field of the PDI Control Register.

Table 6 PDI MODE select

PDI_SEL	PDI_MODE	Direct Map
0x04	Digital IO	
0x05	Beckhoff SPI	Yes
0x90	HBI Multiplexed 1 Phase 8-bit	Yes
0x91	HBI Multiplexed 1 Phase 16-bit	Yes
0x92	HBI Multiplexed 2 Phase 8-bit	Yes
0x93	HBI Multiplexed 2 Phase 16-bit	Yes

2.4.2. MII TX Shift

Phase shift (0/10/20/30ns) of MII TX signals (MII_TXEN, MII_TXD[3:0]) can be attained via the TX_SHI[x] signals. TX Shift settings are explained in Table below. It is recommended to support all TX_SHI[1:0] configurations by hardware options to enable later adjustments.

Table 7 TX SHIFT

TR8253	TR8253L	Config signal	Register	Description
2	16	TX_SHI[0]	0x0E03[1]	TX SHIFT 00 = MII TX signals not delayed 01 = MII TX signals delayed by 10 ns 10 = MII TX signals delayed by 20 ns 11 = MII TX signals delayed by 30 ns
1	15	TX_SHI[1]	0x0E03[2]	

2.4.3. CLK25M Output Enable

A 25MHz clock for external device, e.g. MCU, can be made available by the TR8253/TR8253L on CLK25M pin if it is pulled-up.

Table 8 CLK25 Enable

TR8253	TR8253L	Config signal	Register	Description
45	8	CLK25_EN	0x0E03[5]	CLK25M Enable 0 = disable 1 = enable, CLK25M is 25 MHz clock output

2.4.4. Link Polarity

Ethernet PHYs signal a 100 Mbit/s Full Duplex Link to the TR8253/TR8253L by asserting LINK_MII. The polarity can be selected with LINKPOL.

Table 9 PHY Address Offset

TR8253	TR8253L	Config signal	Register	Description
62	29	LINKPOL	0x0E03[3]	Link Polarity 0 = LINK_MII is active low 1 = LINK_MII is active high

2.4.1. LED Polarity

LEDPOLx determines the polarity of LEDs on corresponding Pins.

Table 10 LED Polarity

TR8253	TR8253L	Config signal	Register	Description
36,14, 62,37,46	48,46 29,45,9	LEDPOLx		LED polarity 0 = LED is active high 1 = LED is active low

2.4.2. SII EEPROM Size

EEPROM_SIZE determines the size of the EEPROM (and the number of I²C address bytes). EEPROM_SIZE is sampled at the beginning of the EEPROM access.

Table 11 SII EEPROM Size

TR8253	TR8253L	Config signal	Register	Description
37	45	EEPROM_SIZE	0x0502[7]	EEPROM Size 0 = 1 address byte (1 Kbit to 16 Kbit EEPROM) 1 = 2 address bytes (32 Kbit to 4 Mbit EEPROM)

2.5. General Pins

Table 12 General pins

TR8253	TR8253L	Signal	I/O
41	1	OSC_IN	I _A
42	2	OSC_OUT	O _A
49	11	RESET#	I _{PU}
15	41	TESTMODE	I _{PD}
38	43	EEPROM_CLK	IO _{PU}
39	42	EEPROM_DATA	IO _{PU}
43	34	SYNC0/LATCH0	IO
44	18	SYNC1/LATCH1	IO
37	45	RUNLED	IO _{PD}
46	9	ERRLED	IO
45	8	CLK25M	IO _{PU}
36	48	LINKACT0	IO _{PU}
14	46	LINKACT1	IO _{PU}
62	29	LINKACT2 ⁽¹⁾	IO

NOTE1:LINKACT2 is valid only when external MII is activated by Chip mode selection.

OSC_IN

Connection to external crystal or oscillator input (25 MHz). An oscillator as the clock source for both TR8253/TR8253L and external PHY. The 25 MHz clock source should have an initial accuracy of 25ppm or better.

OSC_OUT

Connection to external crystal.

OSC_OUT should be left open if an oscillator is connected to OSC_IN.

RESET#

The open collector RESET input/output (active low) signals the reset state of TR8253/TR8253L. The reset state is entered at power-on, if the power supply is to low, or if a reset was initiated using the reset register 0x0040. TR8253/TR8253L also enters reset state if RESET# pin is held low by external devices

TESTMODE

Reserved for testing, should be connected to GND.

EEPROM_CLK

EEPROM I²C clock signal.

EEPROM_DATA

EEPROM I²C data signal.

SYNC0/LATCH0, SYNC1/LATCH1

Distributed Clocks SyncSignal output or LatchSignal input, depending on SII EEPROM configuration. SYNC/LATCH signals are not driven (high impedance) until the EEPROM is loaded

RUNLED

SII EEPROM Size configuration (either 1 Kbit-16 Kbit or 32 KBit-4 Mbit) sampled at the beginning of the EEPROM access. Otherwise RUN LED signal. RUN is active high if pin is pulled down, and active low if pin is pulled up.

STATE_RUNLED

This pin is the STATE_RUN LED output and is equal to the RUN LED combined with the negation of the ERR LED. It can be used to control the RUN side of a bi-color RUN/ERR LED.

ERRLED

This pin is the ERR LED output and is controlled by the ESC.

This pin is configured to be an open-drain/opensource output. The default choice of open-drain vs. open-source as well as the default polarity of this pin depends upon the strap value sampled at reset.

CLK25M

A 25MHz clock for external device, e.g. MCU, can be made available by the TR8253/TR8253L on this pin.

LINKACT0, LINKACKT1, LINKACK2

Link/Activity LEDs output (off=no link, on=link without activity, blinking=link and activity) for physical port x. LINKACT[x] is active high if pin is pulled down, and active low if pin is pulled up.

LINKACT0 outputs for physical phyA, LINKACK1 for physical phyB and LINKACK2 for external PHY.

2.6. Ethernet PHY and MII Pins

The TR8253 pin out is optimized in order to achieve an optimum of size and features. To obtain this, there is several pins where either communication or PDI functionality can be assigned to. Number and type of the communication ports might reduce/exclude one or more PDI possibilities.

The physical communication ports are numbered from port 0 to port 2.

2.6.1. MDI Physical Port

TR8253/TR8253L contains PHYs A and B, which are identical in functionality. PHY A connects to either port 0 or 2 of the EtherCAT Core. PHY B connects to port 1 of the EtherCAT core. These PHYs interface with their respective MAC via an internal MII interface.

The PHYs comply with the IEEE 802.3 Physical Layer for Twisted Pair Ethernet specification and can be configured for full/half duplex 100 Mbps (100BASE-TX) or 10 Mbps (10BASE-T) Ethernet operation. However, only full duplex, 100BASE-TX operation is used for EtherCAT. All PHY registers follow the IEEE 802.3 (clause 22.2.4) specified MII management register set and are fully configurable.

Each individual PHY is assigned a default PHY address. The address for PHY A is set to 0 or 2, based on the device mode. The address for PHY B is fixed to 1.

Table 13 MDI Physical Port

TR8253	TR8253L	Signal	Description	Type
17	53	PHYA_TXP	PhyA MDI Transmit/receive differential pair 0	IO _A
18	52	PHYA_TXN	PhyA MDI Transmit/receive differential pair 0	IO _A
19	55	PHYA_RXP	PhyA MDI Transmit/receive differential pair 1	IO _A
20	54	PHYA_RXN	PhyA MDI Transmit/receive differential pair 1	IO _A
26	62	PHYB_TXP	PhyB MDI Transmit/receive differential pair 0	IO _A
27	63	PHYB_TXN	PhyB MDI Transmit/receive differential pair 0	IO _A
28	60	PHYB_RXP	PhyB MDI Transmit/receive differential pair 1	IO _A
29	61	PHYB_RXN	PhyB MDI Transmit/receive differential pair 1	IO _A
22	-	PHYA_RBIAS ⁽¹⁾	PhyA Reference Bias. Connect a 12kΩ resistor with 1% accuracy to ground.	IO _A
31	-	PHYB_RBIAS ⁽¹⁾	PhyB Reference Bias. Connect a 12kΩ resistor with 1% accuracy to ground.	IO _A
-	57	PHY_RBIAS ⁽²⁾	Reference Bias for PhyA and PhyB. Connect a 12kΩ resistor with 1% accuracy to ground.	IO _A

NOTE1:For TR8253, Reference Bias resistors for PhyA and PhyB are connected to separated pins.

NOTE2:For TR8253L, Reference Bias resistor for PhyA and PhyB share one pin.

2.6.2. MII Physical Port

Table 14 MII Physical Port

TR8253	TR8253L	Signal	Description	Type
5	23	MIITXEN	MIITransmit enable output	O

4	22	MII_TXD0	MII transmit data bit0	O
3	21	MII_TXD1	MII transmit data bit1	O
2	16	MII_TXD2	MII transmit data bit2	O
1	15	MII_TXD3	MII transmit data bit3	O
6	33	MII_RXER	MII receive error	I
7	36	MII_RXCLK	MII Receive Clock	I
8	26	MII_RXDV	MII receive data valid.	I
9	27	MII_RXD0	MII receive data bit0	I
10	28	MII_RXD1	MII receive data bit1	I
11	30	MII_RXD2	MII receive data bit2	I
12	31	MII_RXD3	MII receive data bit3	I
58	49	MII_LINK	MII link status	I
60	39	MII_MDC	PHY Management Interface clock	O
59	40	MII_MDIO ⁽¹⁾	PHY Management Interface Data	IO
61	25	MII_CLK25 ⁽²⁾	25 MHz clock output for external PHY.	O

NOTE1: MII_MDIO must have a pull-up resistor outside(4.7 kΩ recommended for ESCs)

NOTE2: The MII_CLK25 pins provide a clock signal – if configured – during external or ECAT reset, clock output is only turned off during power-on reset.

2.7. PDI Signal Pins

The PDI signal pinout depends on the selected PDI (SII EEPROM). The PDI selection and PDI signal pinout is subject to restrictions introduced by the port configuration. Digital I/O and SPI PDI are available in any configuration – although the I/O width can be reduced depending on the configuration.

The function of PDI signals depends on the configuration stored in the device SII EEPROM. PDI signals are not driven (high impedance) until the EEPROM is loaded. PDI signals are not driven (high impedance) if no PDI is configured (PDI Control register 0x0140=0x00).

The Process Data Interface (PDI) used by the device is indicated via the PDI Control Register. The available PDIs are:

04h: Digital I/O PDI

05h: Beckhoff SPI Interface

83h: SQI Interface

90h, 92h: Host Bus Interface (HBI)

(8-bit, HBI Multiplexed 1/2 Phase, EtherCAT Direct Map)

91h, 93h: Host Bus Interface (HBI)

(16-bit, HBI Multiplexed 1/2 Phase, EtherCAT Direct Map)

94h: Host Bus Interface (HBI)

(8-bit, HBI Indexed, EtherCAT Direct Map)

95h: Host Bus Interface (HBI)

(16-bit, HBI Indexed, EtherCAT Direct Map)

The PDI Control Register is initialized from the contents of EEPROM.

The configuration of the enabled PDI is controlled via the PDI Configuration Register and Extended PDI Configuration Register.

2.7.1. Digital I/O Pins

The Digital I/O PDI provides 16 configurable digital I/Os (DIGIO[15:0]) to be used for simple systems without a host controller. The Digital I/O Output Data Register is used to control the output values, while the Digital I/O Input Data Register is used to read the input values. Each 2-bit pair of the digital I/Os is configurable as an input or output. The direction is selected by the Extended PDI Configuration Register, which is configured via EEPROM.

The Digital I/Os can also be configured to bi-directional mode, where the outputs are driven and latched externally and then released so that the input data can be sampled. Bi-directional operation is selected via the Unidirectional/Bidirectional Mode bit of the PDI Configuration Register. The PDI Configuration Register is initialized from the contents of EEPROM.

Table 15 Digital I/O Interface

TR8253	TR8253L	Signal	Description	Type
54	36	DIGIO0	Digital channel 0	IO
56	39	DIGIO1	Digital channel 1	IO
57	40	DIGIO2	Digital channel 2	IO
58	49	DIGIO3	Digital channel 3	IO
59	23	DIGIO4	Digital channel 4	IO
60	22	DIGIO5	Digital channel 5	IO
61	21	DIGIO6	Digital channel 6	IO
62	16	DIGIO7	Digital channel 7	IO
1	15	DIGIO8	Digital channel8	IO
2	33	DIGIO9	Digital channel9	IO
3	29	DIGIO10	Digital channel10	IO
4	26	DIGIO11	Digital channel11	IO
5	27	DIGIO12	Digital channel12	IO
6	28	DIGIO13	Digital channel13	IO
7	30	DIGIO14	Digital channel14	IO
8	31	DIGIO15	Digital channel15	IO
9	12	SOF	Start of Frame	O
10	13	EOF	End of Frame	O
48	19	LATCH_IN	External data latch signal	I
50	17	WD_STATE	Watchdog State	O
51	35	WD_TRIG	Watchdog Trigger	O
52	50	OUTVALID	Output data is valid/Output event	O
53	25	OE_EXT	Output Enable	I

2.7.2. SPI/SQI Pins

This interface is used for systems with a host controller that provides a compatible SPI/SQI host interface. It supports EtherCAT Direct Mapped mode.

The values in the PDI Configuration Register and the Extended PDI Configuration Register reflect the value from EEPROM. The value in the PDI Configuration Register is used to configure the SPI/SQI interface.

The PDI Configuration Register and Extended PDI Configuration Register are initialized from the contents of the EEPROM.

Table 16 SPI/SQI Interface pins

TR8253	TR8253L	Signal	Description	Type
48	44	IRQ	Interrupt output to host controller	O
50	50	SPI_CS#	SPI/SQI chip select, active low	I
51	19	SPI_SCK	SPI/SQI Serial clock	I
52	17	SPI_MOSI/SIO0	SPI master output slave input/SQI D0	IO
53	13	SPI_MISO/SIO1	SPI master input slave output/SQI D1	IO
9	12	SIO2	SQI D2	IO
10	35	SIO3	SQI D3	IO

2.7.3. GPIO Pins

The EtherCAT Core provides 16 General Purpose Inputs (GPI[15:0]) and 16 General Purpose Outputs (GPO[15:0]) The General Purpose Output Register is used to control the output value. The General Purpose Input Register is used to read the input value.

Each 2-bit pair is configurable as input, push-pull output or open-drain output. The direction and buffer type are determined by the Extended PDI Configuration Register. Bits 7:0 control the direction of the pairs (bit 0 for GPIO[1:0], bit 1 for GPIO[3:2], etc.). A value of 1 selects the output direction. Bits 15:8 control the output type (bit 8 for GPIO[1:0], bit 9 for GPIO[3:2], etc.). A value of 1 selects the open-drain. The Extended PDI Configuration Register is initialized from the contents of the EEPROM.

Table 17 GPIO pins

TR8253	TR8253L	Signal	Description	Type
54	36	GPIO0	GPIO channel0	IO
56	39	GPIO1	GPIO channel1	IO
57	40	GPIO2	GPIO channel2	IO
58	49	GPIO3	GPIO channel3	IO
59	23	GPIO4	GPIO channel4	IO
60	22	GPIO5	GPIO channel5	IO
61	21	GPIO6	GPIO channel6	IO
62	16	GPIO7	GPIO channel7	IO
1	15	GPIO8	GPIO channel8	IO
2	33	GPIO9	GPIO channel9	IO
3	29	GPIO10	GPIO channel10	IO
4	26	GPIO11	GPIO channel11	IO
5	27	GPIO12	GPIO channel12	IO
6	28	GPIO13	GPIO channel13	IO
7	30	GPIO14	GPIO channel14	IO
8	31	GPIO15	GPIO channel15	IO

2.7.4. HBI μ Controller Pins

This interface is used for μ Controller systems with a Host Bus Interface. It supports EtherCAT Direct Mapped mode.

The values in the PDI Configuration Register and the Extended PDI Configuration Register reflect the value from EEPROM. The values in the PDI Configuration Register are used to configure the HBI.

The values in the Extended PDI Configuration Register are used to further configure the HBI while in EtherCAT Direct Mapped mode.

The PDI Configuration Register and Extended PDI Configuration Register are initialized from the contents of the EEPROM.

Table 18 HBI Multiplexed/Indexed Interface pins

TR8253	TR8253L	Signal		Description		Type
		Multiplexed	Indexed	Multiplexed	Indexed	
48	44	IRQ		Interrupt output to host controller		O
50	28	CS		chip select		I
51	30	WR/ENB		Mode0: Write command Mode1: write/ read enable		I
52	31	RD/WR_RD		Mode0: Read command Mode1: write/ read direction		I
53	10	WAIT_ACK		Wait acknowledge		O
54	17	AD0	D0	Address Data bus bit0	Data bus bit0	IO
56	13	AD1	D1	Address Data bus bit1	Data bus bit1	IO
57	12	AD2	D2	Address Data bus bit2	Data bus bit2	IO
58	35	AD3	D3	Address Data bus bit3	Data bus bit3	IO
59	49	AD4	D4	Address Data bus bit4	Data bus bit4	IO
60	50	AD5	D5	Address Data bus bit5	Data bus bit5	IO
61	36	AD6	D6	Address Data bus bit6	Data bus bit6	IO
62	39	AD7	D7	Address Data bus bit7	Data bus bit7	IO
1	40	AD8	D8	Address Data bus bit8	Data bus bit8	IO
2	19	AD9	D9	Address Data bus bit9	Data bus bit9	IO
3	23	AD10	D10	Address Data bus bit10	Data bus bit10	IO
4	22	AD11	D11	Address Data bus bit11	Data bus bit11	IO
5	21	AD12	D12	Address Data bus bit12	Data bus bit12	IO
6	16	AD13	D13	Address Data bus bit13	Data bus bit13	IO
7	15	AD14	D14	Address Data bus bit14	Data bus bit14	IO
8	33	AD15	A0/D15	Address Data bus bit15	Address bus bit 0/ Data bus bit15	IO
9	25	ALELO	A1	Address lower byte latch enable	Address bus bit 1	I
10	29	ALEHI	A2	Address higher byte latch enable	Address bus bit 2	I
11	26	BE0	A3	Data Byte0 enable	Address bus bit 3	I
12	27	BE1	A4	Data Byte1 enable	Address bus bit 4	I

3. Reference Schematics

3.1. Power supply

TR8253/TR8253L supports single 3.3V Power supply. The internal LDO outputs 1.8V supply for logic core, IO PAD, and PLL.

Recommendation for voltage stabilization capacitors 10uF for VDD_IO, VDD_CORE, VDD_PLL PHYx_DVDD33, PHYx_AVDD33 and PHYx_AVDD12 respectively and 100nF ceramic capacitors for each power pin pair.

For TR8253, 2.2uF stabilization capacitors are recommended for PHYx_REGOUT. Low ESR magnetic beads or 0Ω resistance should be used to isolate the analog power supply pin (PHYx_AVDD12) from the LDO power supply pin (PHYx_REGOUT).

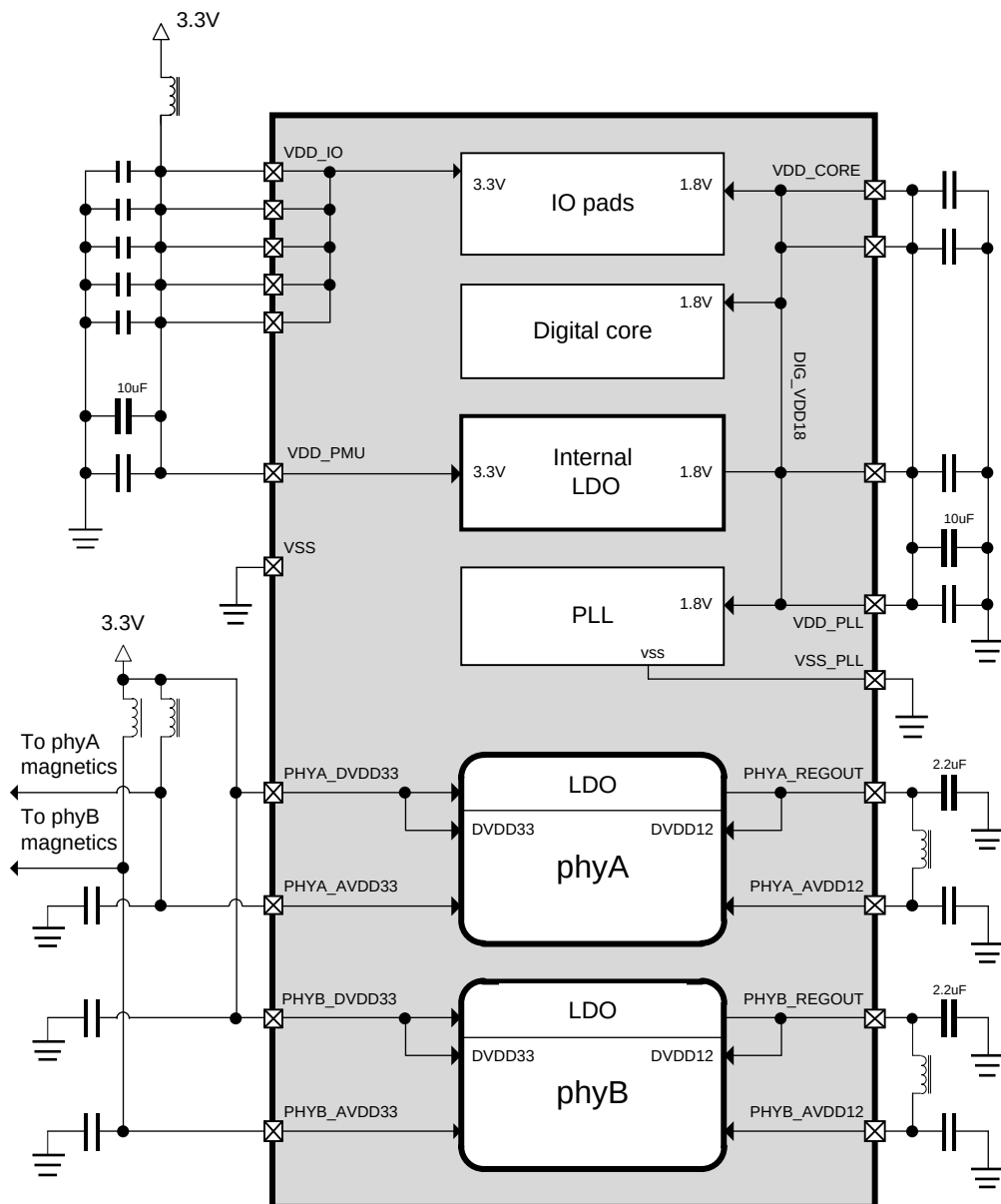


Figure 4 TR8253 Power supply

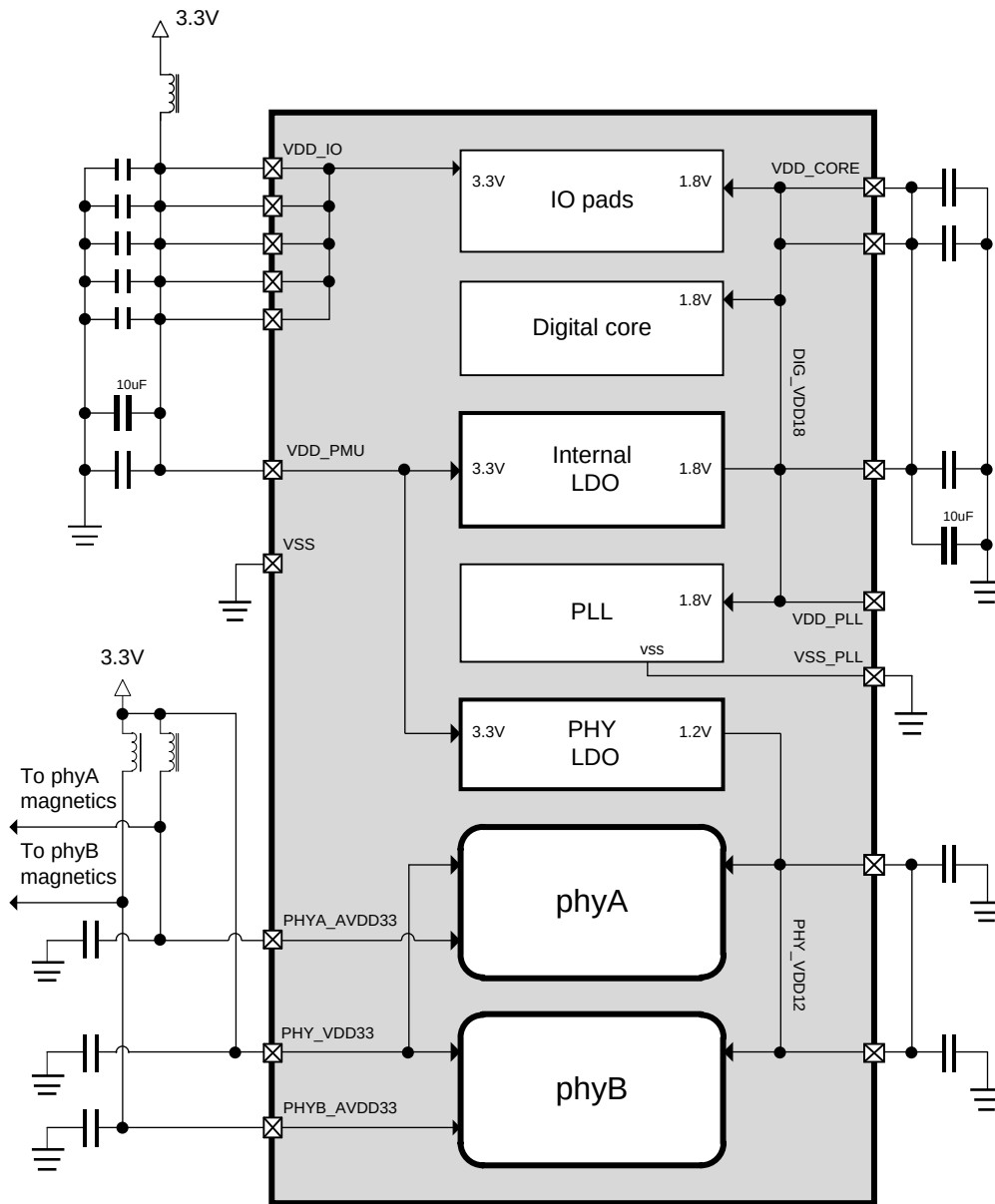


Figure 5 TR8253L Power supply

3.2. Clock source

There are three clock source options for TR8253/TR8253L as shown in following diagram.

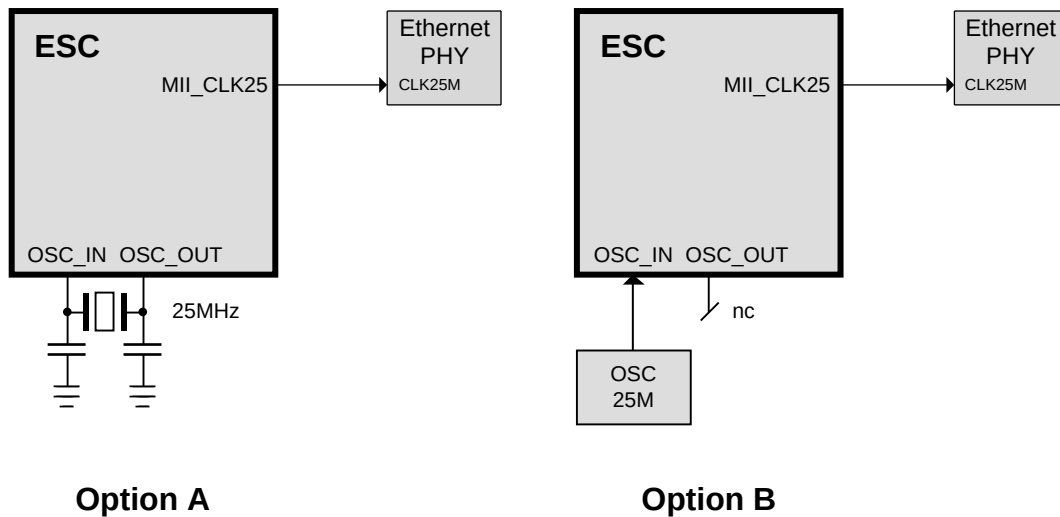


Figure 6 Clock source options

A): Use Quartz crystal as clock source for ESC, and ESC outputs 25MHz clock for Ethernet PHY through pin MII_CLK25.

The value of the load capacitors depends on the load capacitance of the crystal, the pin capacitance C_{OSC} of the ESC pins and the board design (typical 18pF each if $C_L = 16pF$).

See following table for the recommended crystal specifications.

Table 19 Crystal specifications recommendation

Symbol	Parameter	Min	Typ	Max	Unit
f_{osc}	Frequency of crystal	-25ppm	25	+25ppm	MHz
C_L	Load Capacitance			18	pF
C_o	Shunt Capacitance			7	pF
ESR	Equivalent Series Resistance			80	Ω

B): Use Oscillator as clock source for ESC, and ESC outputs 25MHz clock for Ethernet PHY through pin MII_CLK25.

The initial accuracy of the TR8253/TR8253L clock source has to be 25ppm or better.

3.3. External Ethernet PHY

The TR8253/TR8253L is connected with external Ethernet PHYs using the MII interfaces. The MII interfaces of the TR8253/TR8253L are optimized for low processing/forwarding delays by omitting a transmit FIFO. To allow this, the TR8253/TR8253L has additional requirements to Ethernet PHYs, which are easily accomplished by several PHY vendors.

TR8253/TR8253L connects external 100M fast Ethernet PHY signal as following diagram.

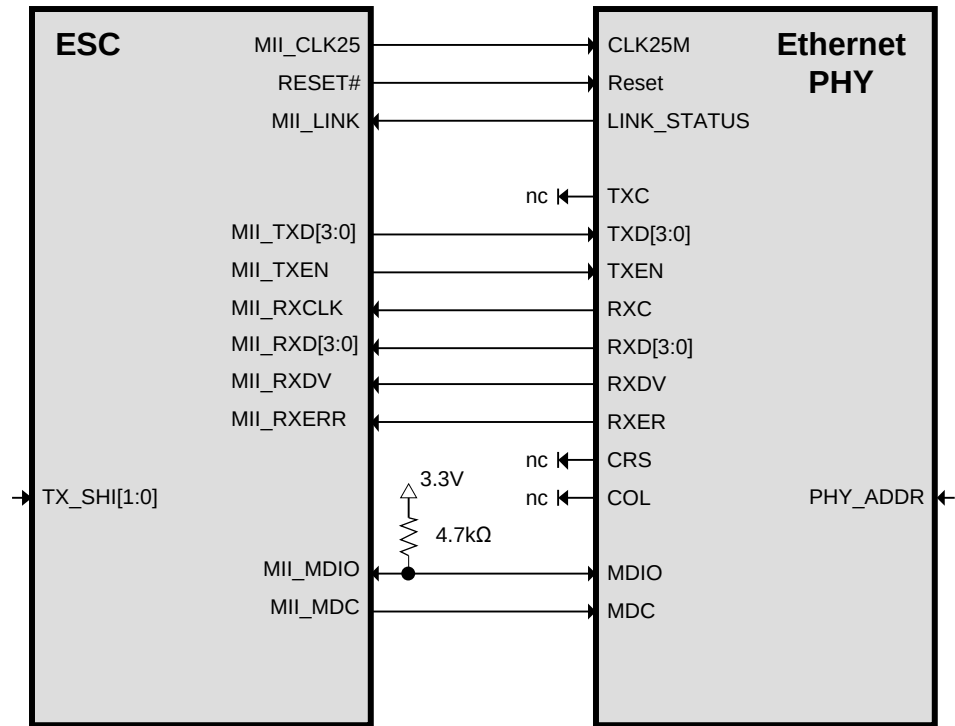


Figure 7 Ethernet PHY connection

3.4. Reset logic

TR8253/TR8253L connects external RESET signal as following diagram.

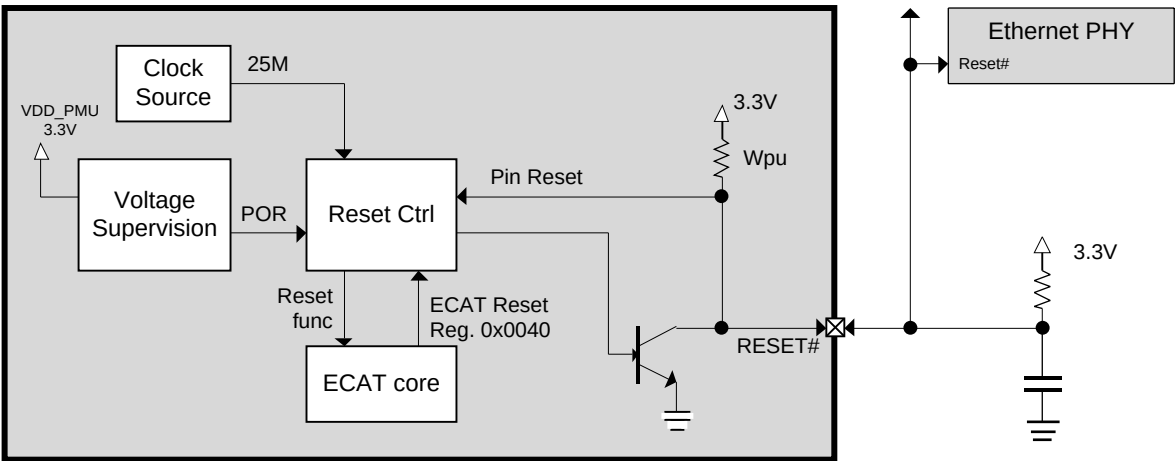


Figure 8 Reset logic

4. Timing Diagrams

4.1. MII Tx Shift timing

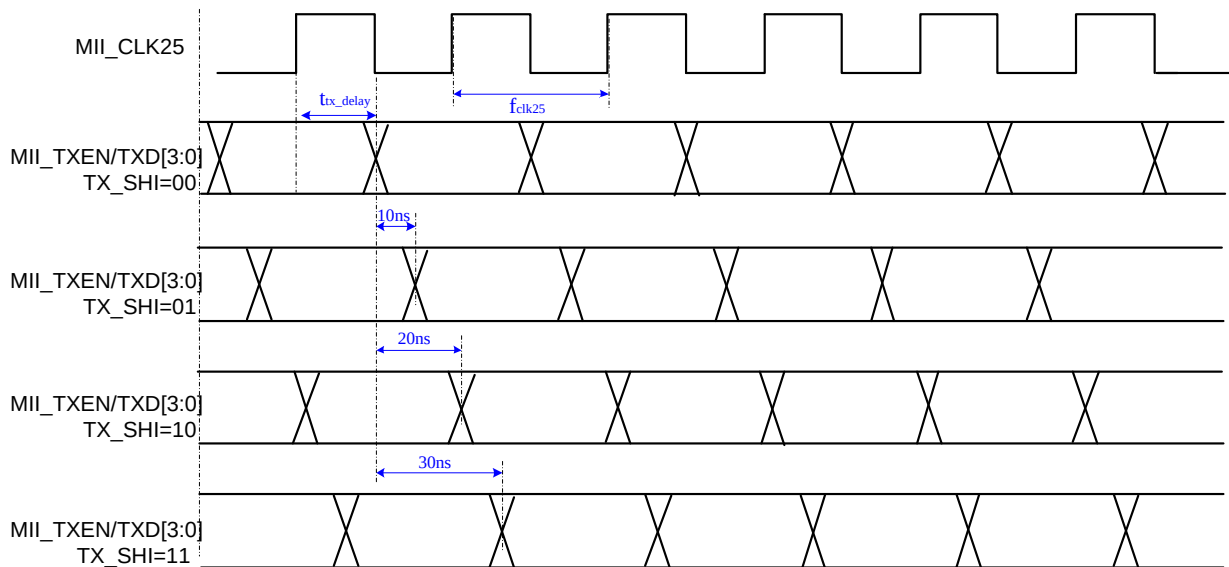


Figure 9 MII Tx Shift Timing

Table 20 MII Tx Shift timing characteristics

Symbol	Parameter	Min	Typ	Max	Unit
f_{clk25}	25 MHz clock source frequency	-25ppm	25	+25ppm	MHz
t_{tx_delay}	MII_TXEN/TXD[3:0] edge (TX_SHI[1:0] = 00) after rising edge of OSC_IN	20		23	ns

4.2. HBI interface timing

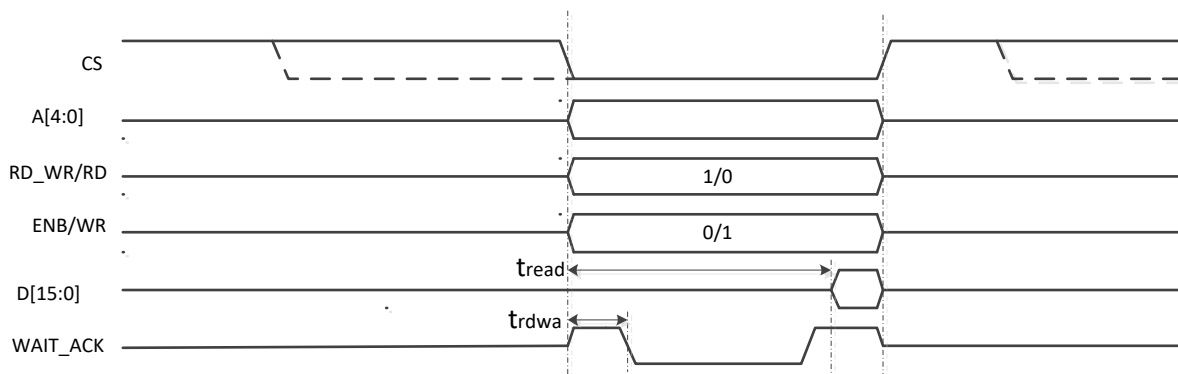


Figure 10 HBI Indexed Addressing Read Timing

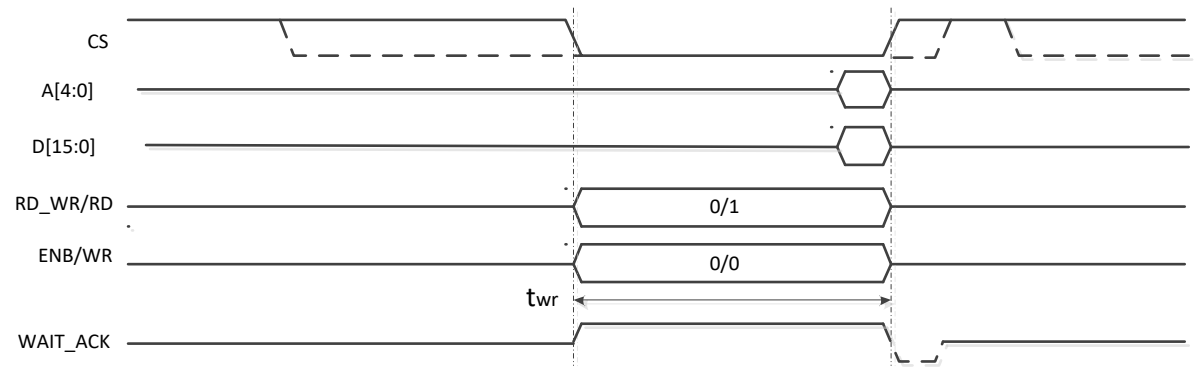


Figure 11 HBI Indexed Addressing Posted Write Timing

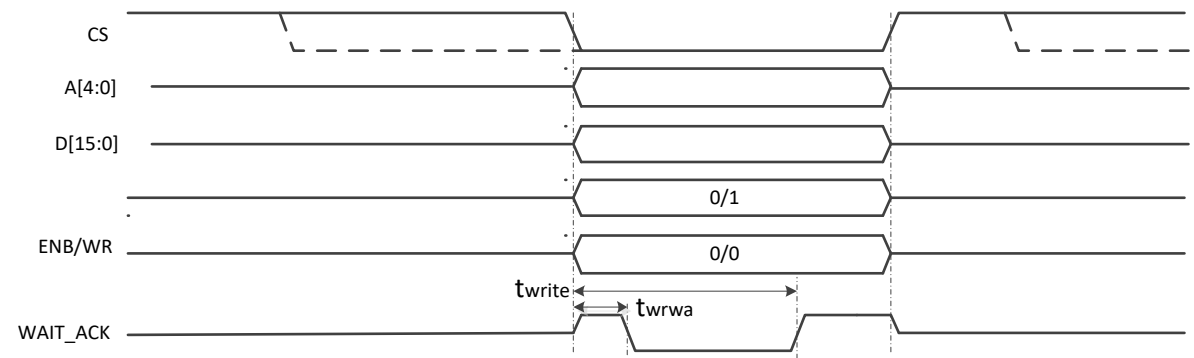


Figure 12 HBI Indexed Addressing Non-Posted Write Timing

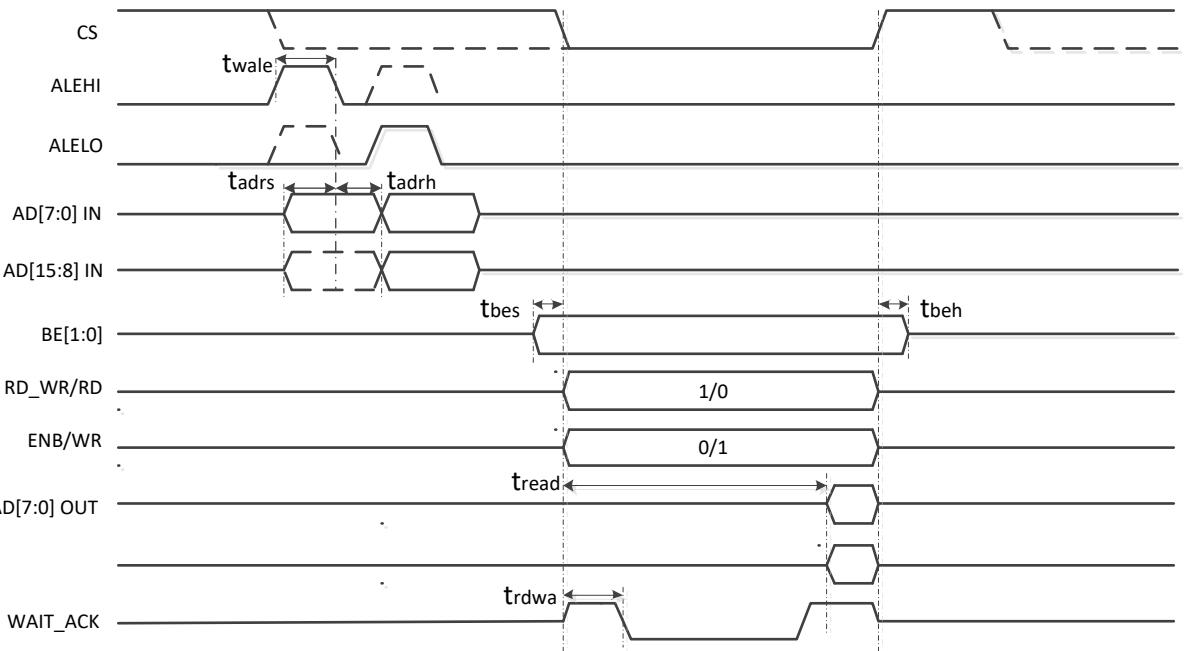


Figure 13 HBI Multiplexed Addressing Read Timing

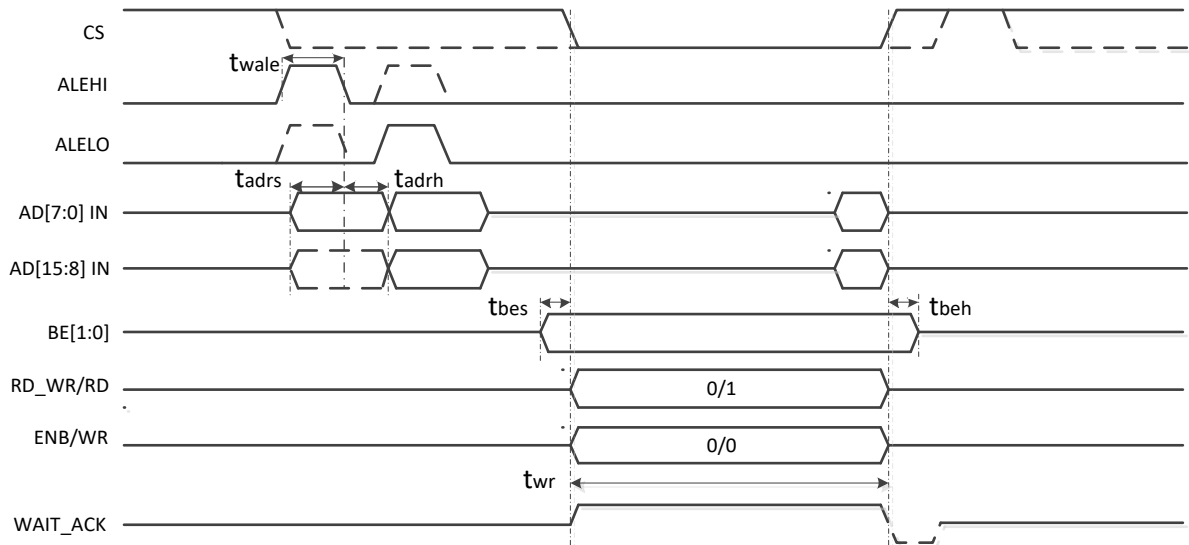


Figure 14 HBI Multiplexed Addressing Posted Write Timing

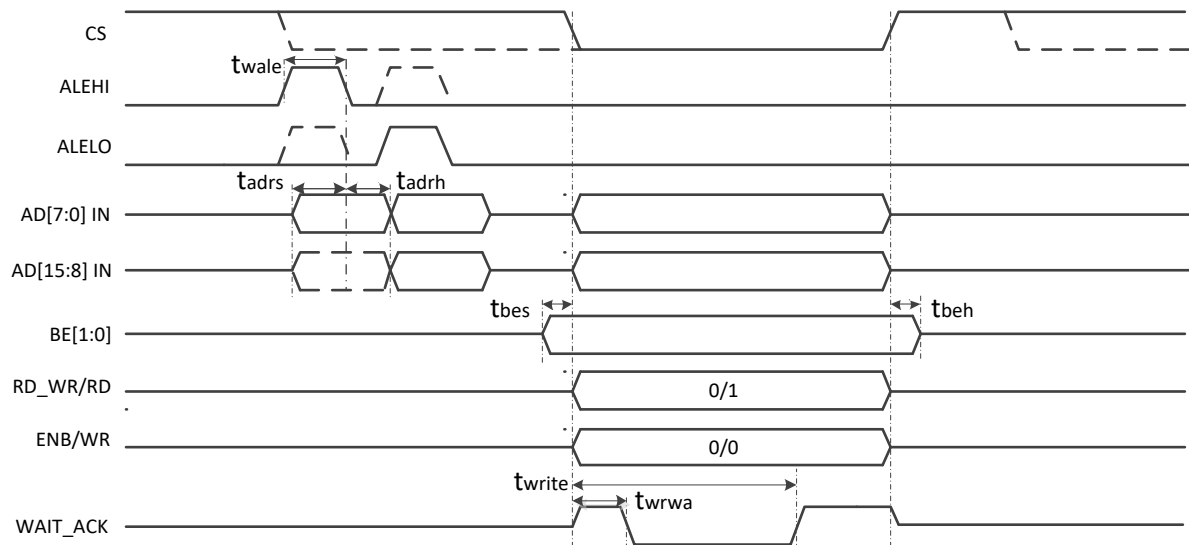


Figure 15 HBI Multiplexed Addressing Non-Posted Write Timing

NOTE: The diagrams in this section show active-low push-pull WAIT_ACK and active-high BE1/0, ALEHI/ALELO, active-low CS, RD, and WR signals. The polarities of these signals are selectable via the HBI WAIT_ACK Polarity, HBI BE1/BE0 Polarity, HBI ALE Polarity, HBI Chip Select Polarity, HBI Read, Read/Write Polarity, and HBI Write, Enable Polarity bits of the PDI Configuration Register and Extended PDI Configuration Register.

Table 21 HBI Timing characteristics

Symbol	Parameter	Min	Typ	Max	Unit
t_{wale}	ALELO, ALEHI Pulse Width	10			ns
t_{ads}	Address Setup to ALELO, ALEHI Inactive	10			ns
t_{adrh}	Address Hold from ALELO, ALEHI Inactive	5			ns
t_{bes}	Byte Enable Setup to WR or ENB Active	1			ns
t_{beh}	Byte Enable Hold from WR or ENB Inactive	0			ns
t_{read}	RD or ENB Active to WAIT_ACK Inactive	150		230	ns
t_{rdwa}	RD or ENB Active to WAIT_ACK Active			10	ns

t_{wr}	WR or ENB Active Time	32	ns
t_{write}	WR or ENB Active to WAIT_ACK Inactive	100	160
t_{wrwa}	WR or ENB Active to WAIT_ACK Active	0	ns

4.3. SPI interface timing

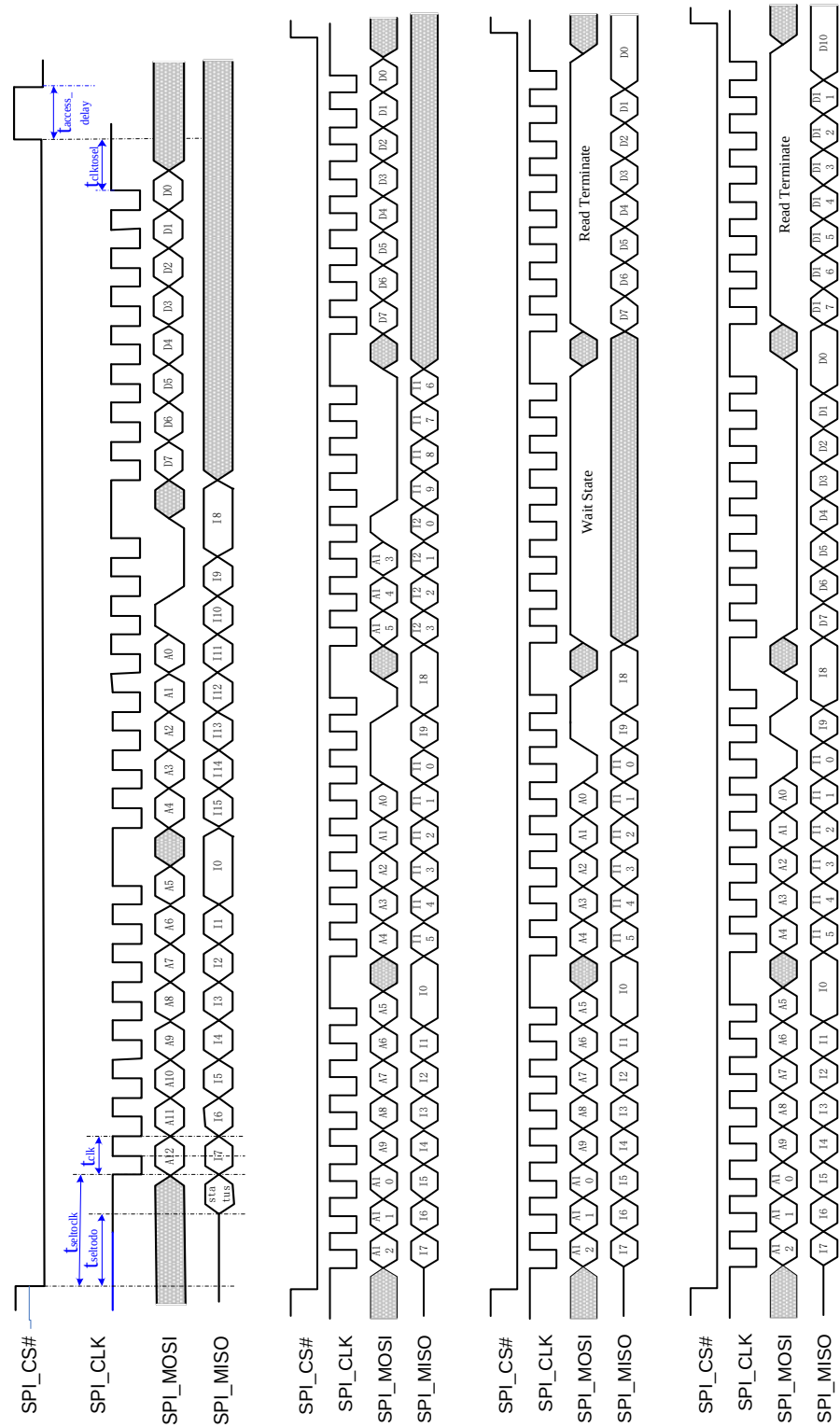
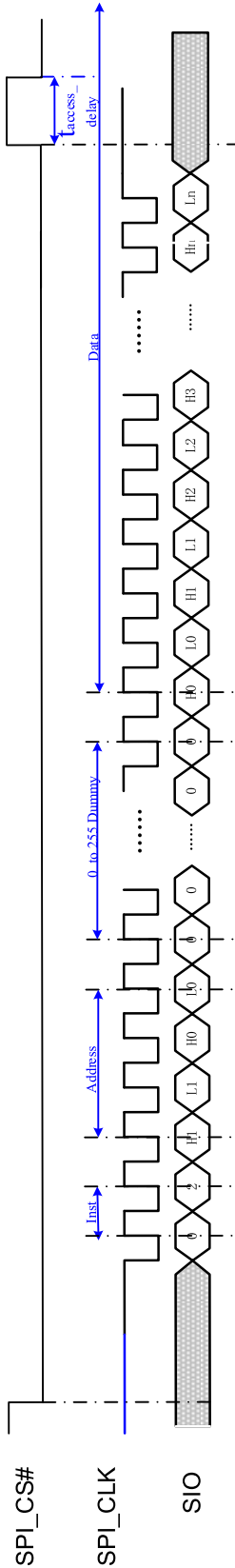


Figure 16 SPI Read Write Timing

4.4. SQI interface timing

WRITE



READ

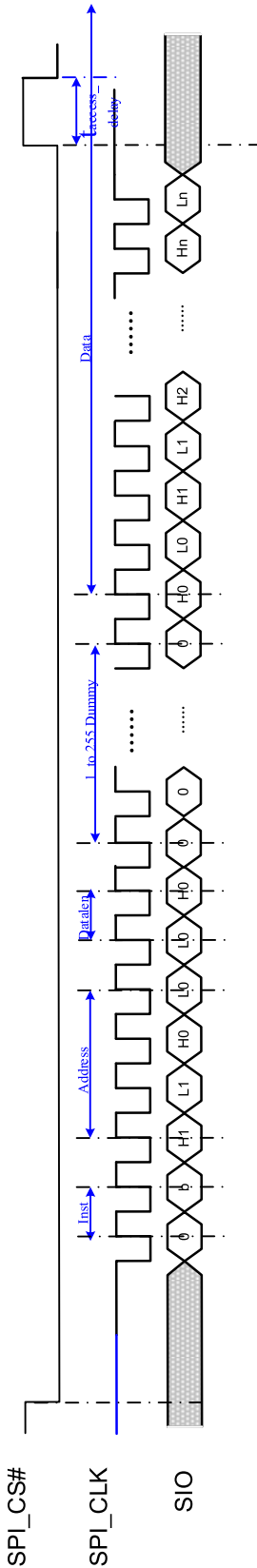


Figure 17 SQI Read Write Timing

Table 22 SPI/SQI Timing characteristics

Symbol	Parameter	Min	Typ	Max	Unit
f_{clk}	SPI_CLK frequency in standard mode			50	MHz
f_{clk_fast}	SPI_CLK frequency in SQI fast mode	40		80	MHz
$t_{seltoclk}$	First SPI_CLK cycle after SPI_CS# asserted	7			ns
$t_{seltodo}$	SPI_MISO/SIO valid after SPI_CLK edge			15	ns
t_{access_delay}	Delay between SPI/SQI accesses	15			ns
$t_{clktsel}$	De-assertion of SPI_CS# after last SPI_CLK cycle	5			ns

4.5. Power up and Reset timing

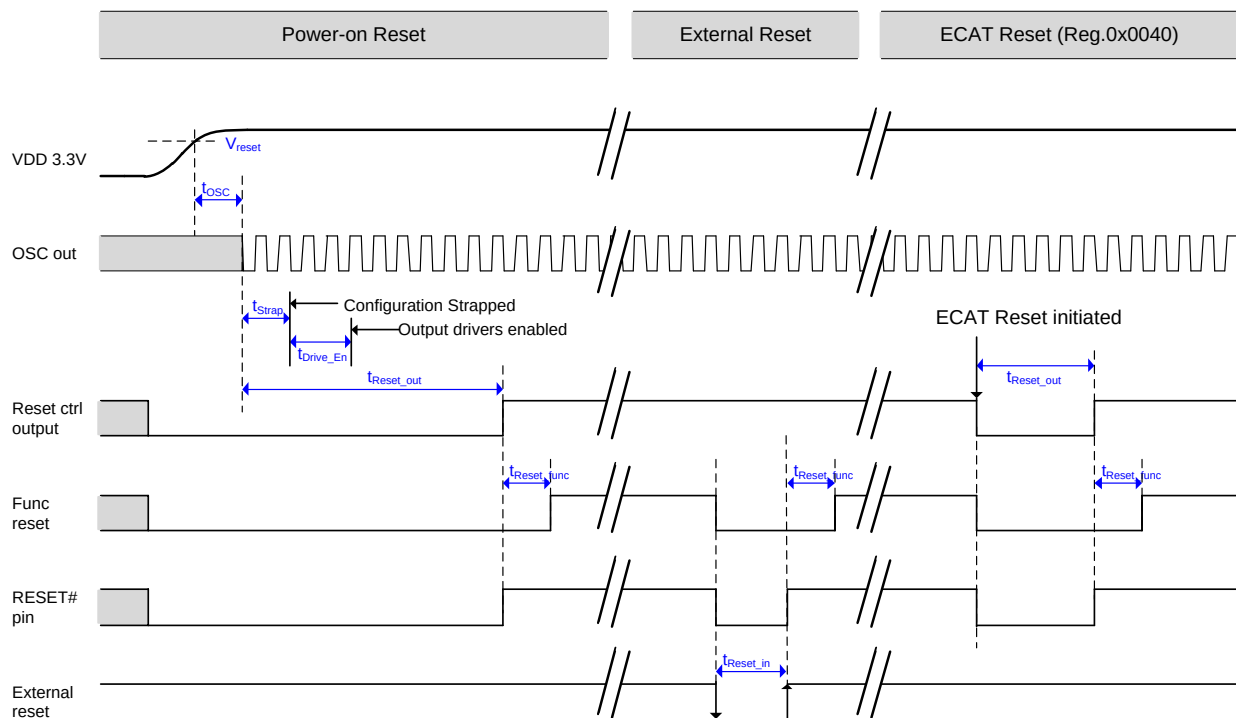


Figure 18 Reset Timing

Table 23 Reset Timing characteristics

Symbol	Parameter	Min	Typ	Max	Unit
t_{osc}	OSC clock stable after 3.3V VDD power is on			10	ms
t_{strap}	Power-on-reset value strap time after OSC clock stable		10		ms
t_{drive_en}	Output drivers enabled after Power-on-reset values strapped (not PDI and not Sync/LatchSignals)		1.2		us
t_{reset_out}	ESC signal RESET# output time	80		100	ms
t_{reset_func}	ESC functional reset release after RESET# signal goes high			50	us
t_{reset_in}	External reset input time	50			ns

5. Electrical Specifications

Table 24 Absolute Maximum Ratings

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{ESD} MDI	ESD protection voltage of MDI Pins (PHYx_TXP/TXN, PHYx_RXP/RXN)	Human body model, according to MIL-STD-883E-3015.7 Class 1			10	kV
V_{ESD} non-MDI	ESD protection voltage except for MDI	Human body model, according to MIL-STD-883E-3015.7 Class 1			6	kV
$T_{storage}$	Storage temperature		-55		125 ⁽¹⁾	°C

Note1: 125°C is only used as bake temperature for not more than 24 hours. Long-term storage (for example, weeks or longer) should be kept at 85°C or lower.

Table 25 Operating Conditions

Symbol	Parameter	Condition	Min	Typ	Max	Unit
VDD PMU	Supply voltage for internal LDO		2.97	3.3	3.63	V
VDD IO	Supply voltage for IO		2.97	3.3	3.63	V
VDD core	Supply voltage for core		1.62	1.8	1.98	V
VDD PLL	Supply voltage for PLL		1.7	1.8	1.9	V
AVDD33	Supply voltage of Analog 3.3V of PHYs		2.97	3.3	3.63	V
DVDD33	Supply voltage of Digital 3.3V of PHYs		2.97	3.3	3.63	V
AVDD12	Supply voltage of Analog 1.2V of PHYs		1.08	1.2	1.32	V
I _{cc}	Supply current for total chip			186 ⁽¹⁾		mA
T_A	Ambient temperature		-40		85	°C
T_J	Maximum Junction Temperature				125	°C

Note1: Typical supply current is measured at ambient temperature 25°C, single 3.3V supply with EtherCAT traffic.

Table 26 DC Characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{IL}	Input Low voltage (not OSC_IN, PHYx_TXN/TXP, PHYx_RXN/RXP)				0.7	V
V_{IH}	Input High voltage (not OSC_IN, PHYx_TXN/TXP, PHYx_RXN/RXP)		2.0		3.6	V
V_{OL}	Output Low voltage				0.4	V
V_{OH}	Output High voltage		2.4			V
R _{wpu}	Weak internal pull-up resistor		90	110	130	kΩ
R _{wpd}	Weak internal pull-down resistor		75	95	115	kΩ
R _{Bias}	Bias resistor for Pin RBIAS		+1%	12	+1%	kΩ

6. Mechanical Dimensions

6.1. TR8253 Package information

TR8253 package type is 64-pin LQFN 9×9 mm, 0.5-mm pitch.

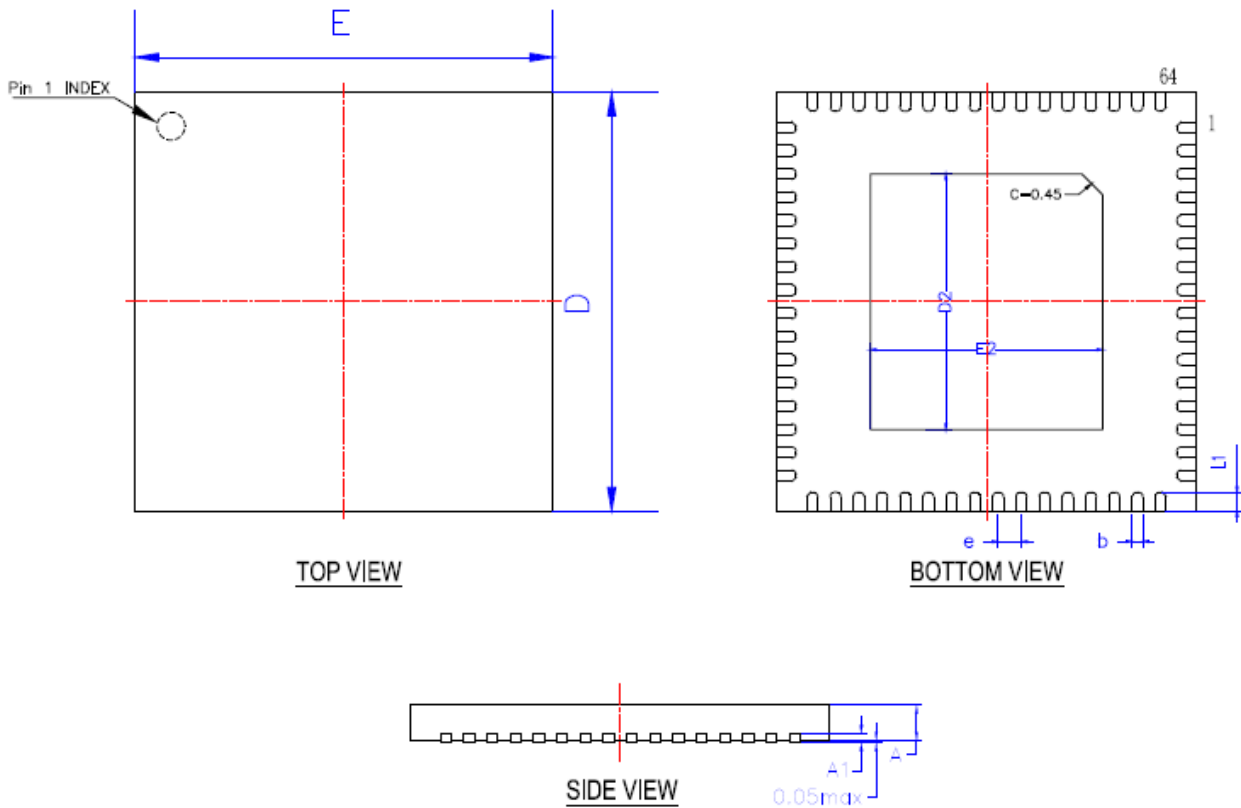


Figure 19 TR8253 Package Outline

Table 27 TR8253 Package Dimensions

Symbol	Min	Nom	Max
A	0.70	0.75	0.80
A1	0.20 BSC		
D	8.90	9.00	9.10
E	8.90	9.00	9.10
D2	5.40	5.50	5.60
E2	4.90	5.00	5.10
e	0.50 BSC		
L1	0.35	0.40	0.45
b	0.18	0.23	0.28

Note1: All dimensions are in millimeters.

6.2. TR8253L Package information

TR8253L package type is 64-pin LGA 9×9 mm, 0.5-mm pitch.

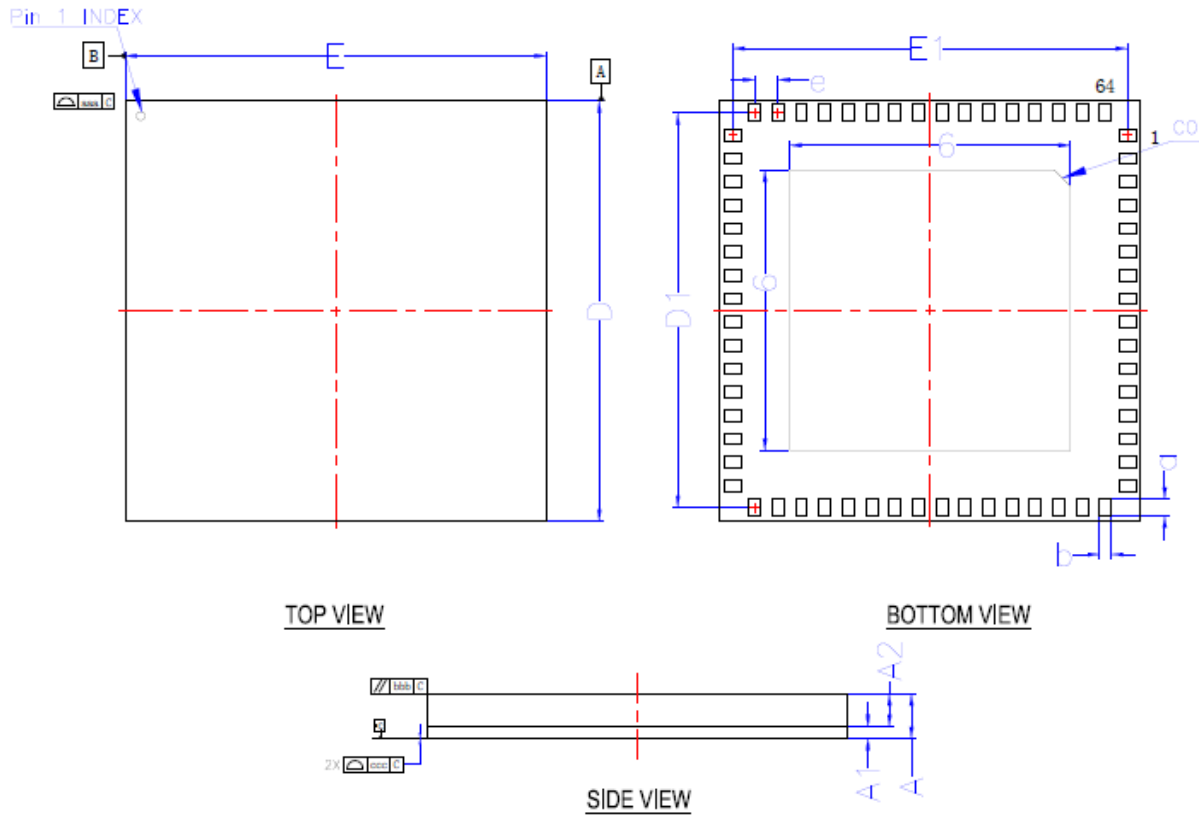


Figure 20 TR8253L Package Outline

Table 28 TR8253L Package Dimensions

Symbol	Min	Nom	Max
A	0.91	0.96	1.01
A1	0.23	0.26	0.29
A2	0.68	0.70	0.72
D	8.90	9.00	9.10
E	8.90	9.00	9.10
D1	8.44 REF		
E1	8.44 REF		
a	0.36 REF		
B	0.25 REF		
e	0.50 REF		
aaa	0.10		
bbb	0.10		
ccc	0.10		

Note1: All dimensions are in millimeters.

Note2: Dimensioning and tolerance per ANSI Y14.5M-1994.

7. Part Order Numbering

7.1. Part Order Numbering

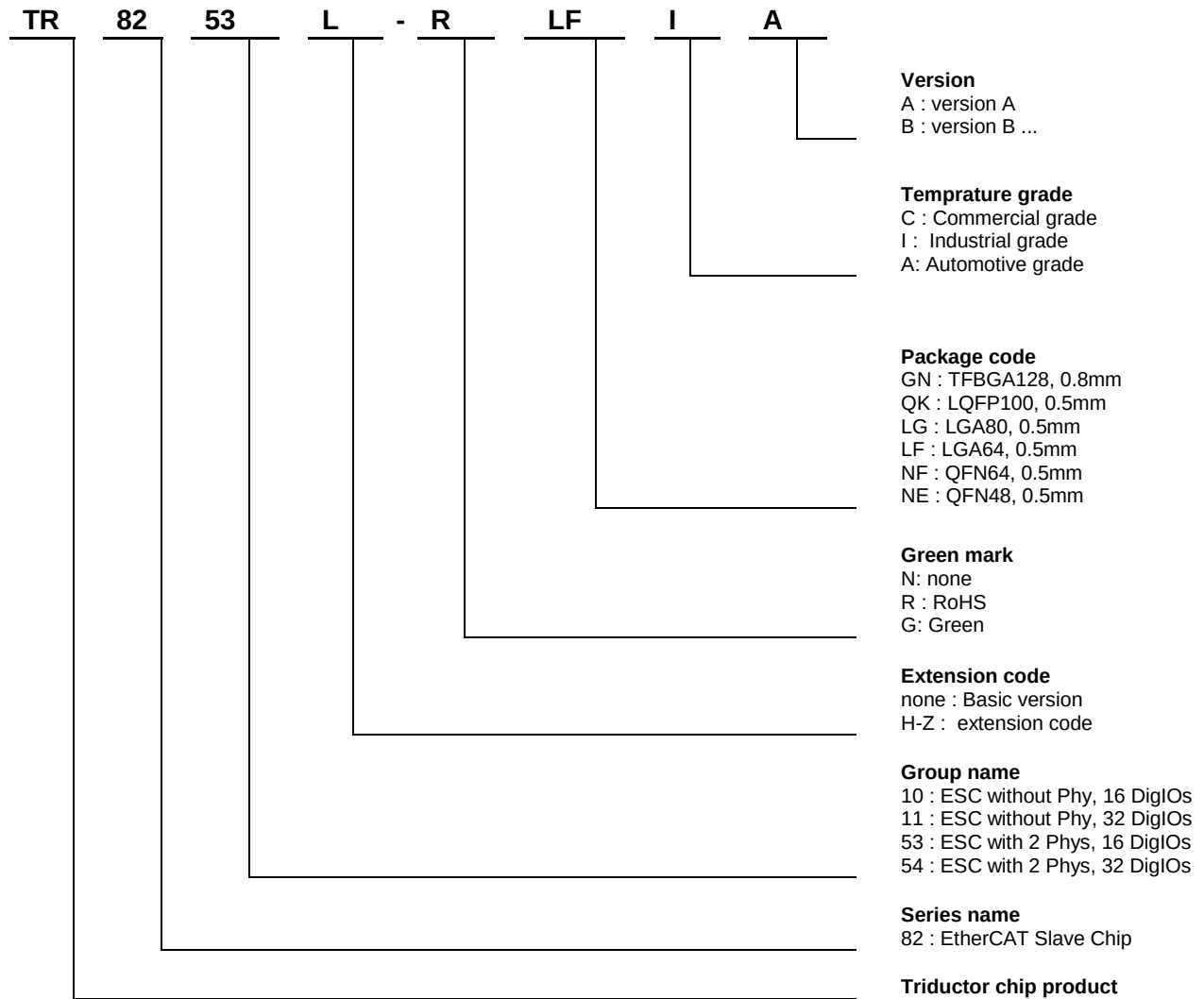


Table 29 Part Order Options

Part Number	Package type	Part Order Number
TR8253	64-pin QFN 9×9 mm, 0.5-mm pitch RoHS, Industrial grade	TR8253-RNFIA
TR8253L	64-pin LGA 9×9 mm, 0.5-mm pitch RoHS, Industrial grade	TR8253L-RLFIA